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(54) **SEAT RECLINING POSITION CONTROL DEVICE**

(71) Applicant: **DENSO CORPORATION**, Kariya (JP)

(72) Inventors: **Ayako Kotani**, Kariya (JP); **Yukari Itou**, Kariya (JP); **Akihiro Hayashi**, Kariya (JP); **Makiko Sugiura**, Kariya (JP); **Kunihiko Chiba**, Kariya (JP)

(73) Assignee: **DENSO CORPORATION**, Kariya (JP)

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Primary Examiner — Khoi H Tran

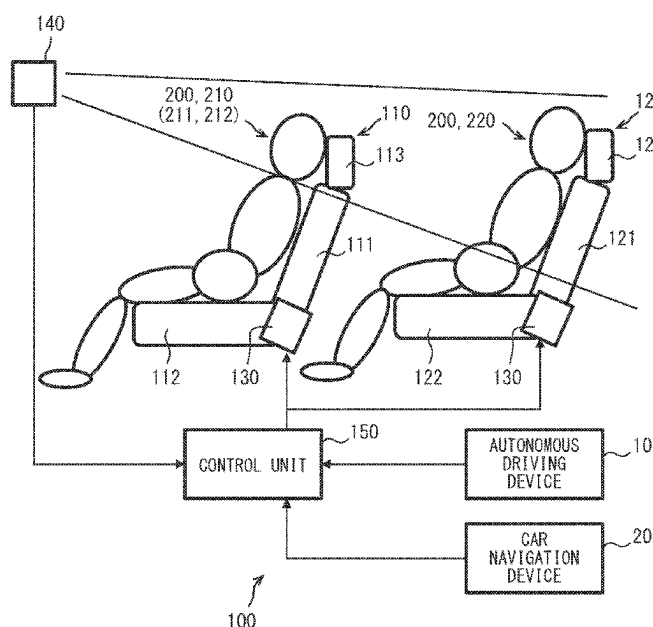
Assistant Examiner — Alyse Tramanh Tran

(74) *Attorney, Agent, or Firm* — Harness, Dickey & Pierce, P.L.C.

(57) **ABSTRACT**

A seat reclining position control device includes: a seat on which an occupant of a vehicle sits; an adjustment part configured to adjust an angle of a backrest of the seat; a sleep state detection unit configured to detect a sleep state of the occupant; a control unit that operates the adjustment unit to control the angle; and a vehicle interior detection unit configured to detect a vehicle interior state of the vehicle. The control unit implements a seat angle control to adjust the angle of the backrest for the occupant in the sleep state based on the vehicle interior state to a suitable sleeping angle suitable for sleeping when the sleep state detection unit detects the sleep state of the occupant.

19 Claims, 7 Drawing Sheets



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B60W 50/16 (2020.01)
G01C 21/26 (2006.01)
 See application file for complete search history.
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- (58) **Field of Classification Search** FOREIGN PATENT DOCUMENTS
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FIG. 1

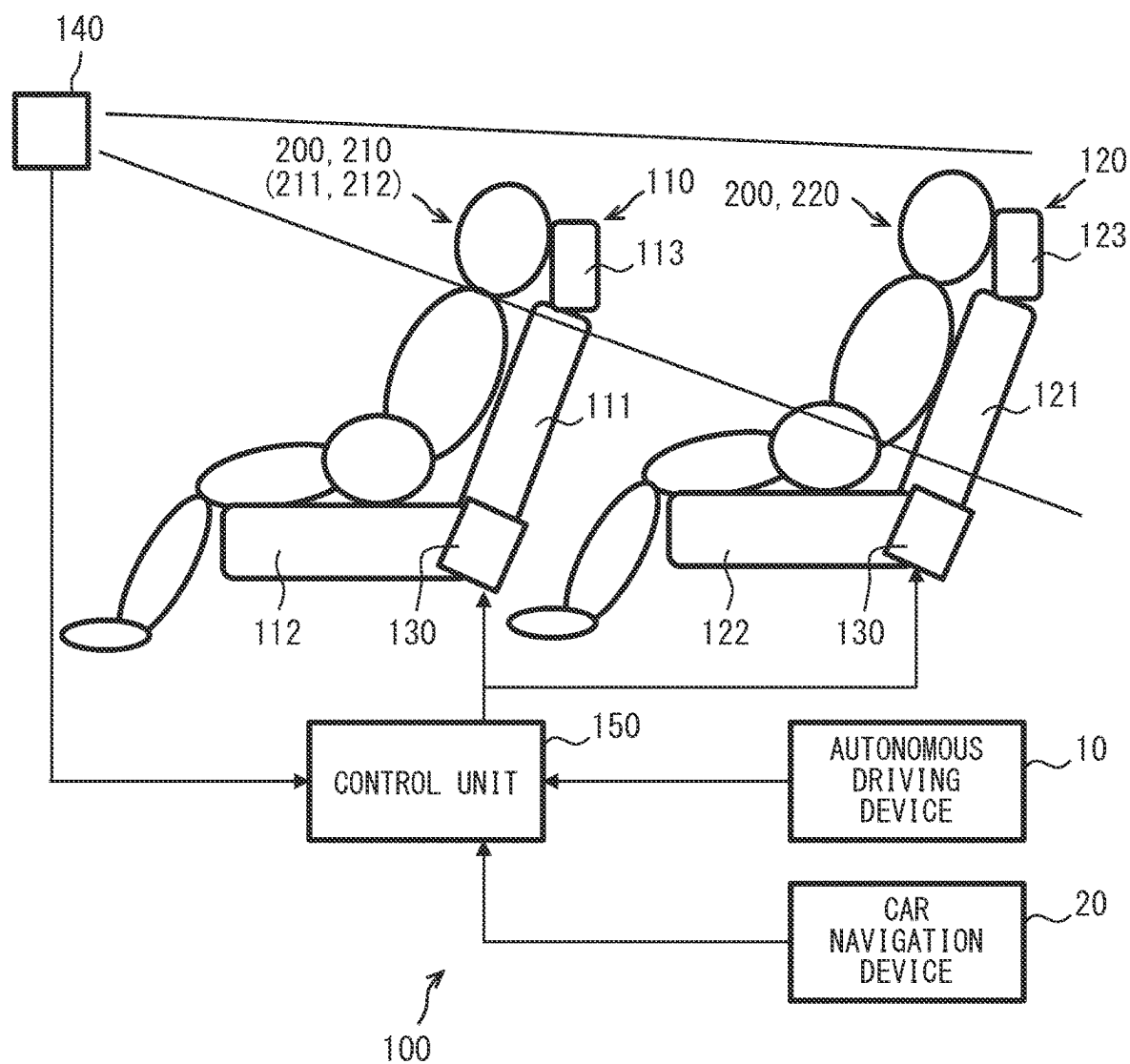


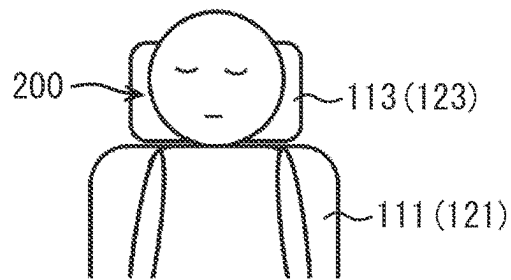
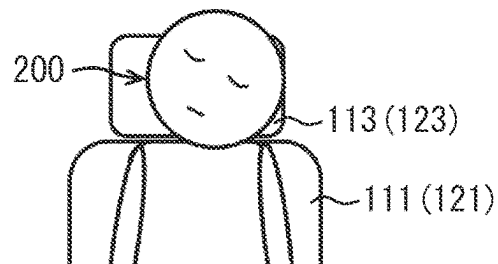
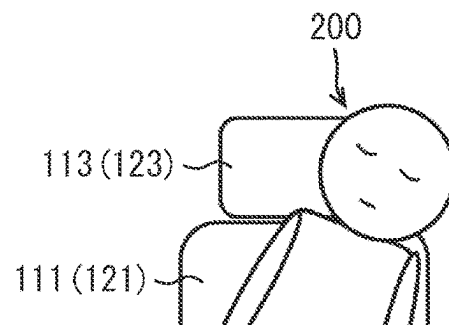
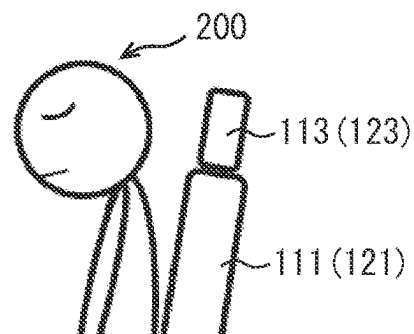
FIG. 2**FIG. 3****FIG. 4****FIG. 5**

FIG. 6

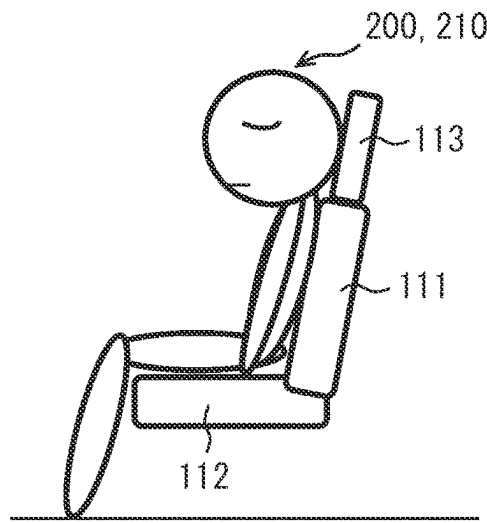


FIG. 7

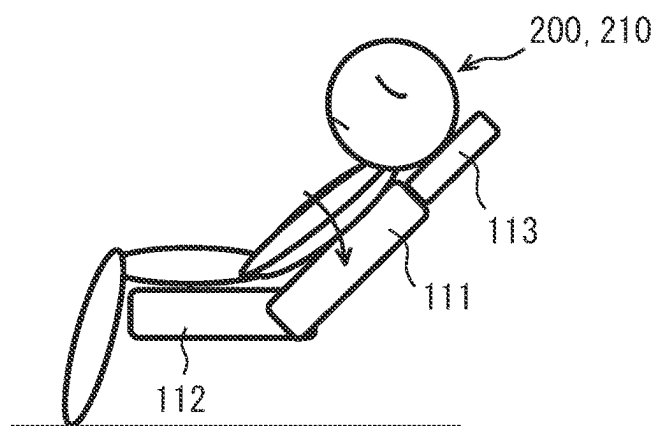


FIG. 8

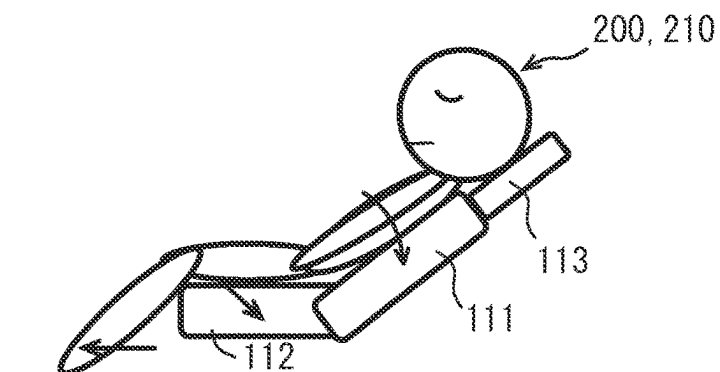


FIG. 9

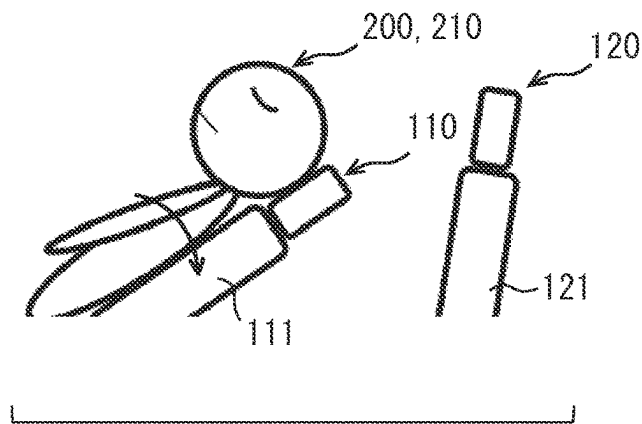


FIG. 10

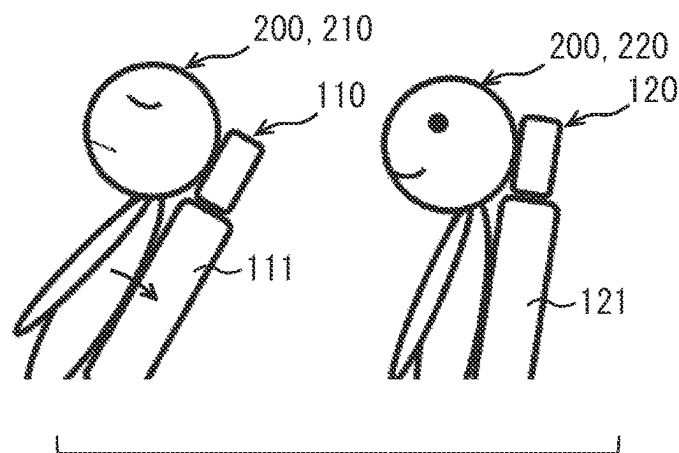


FIG. 11

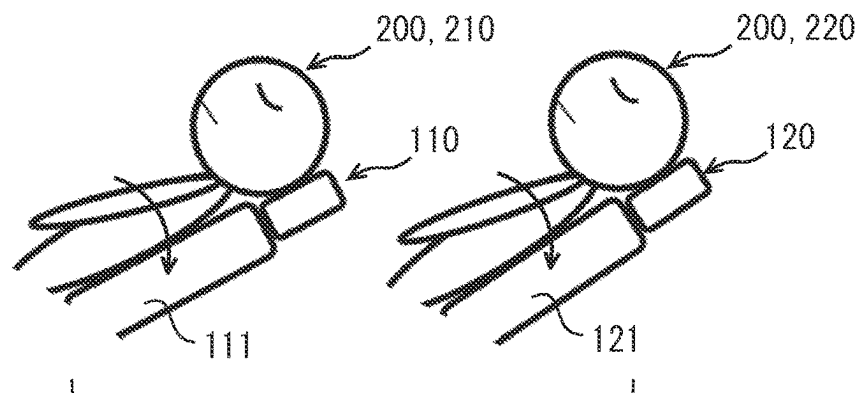


FIG. 12

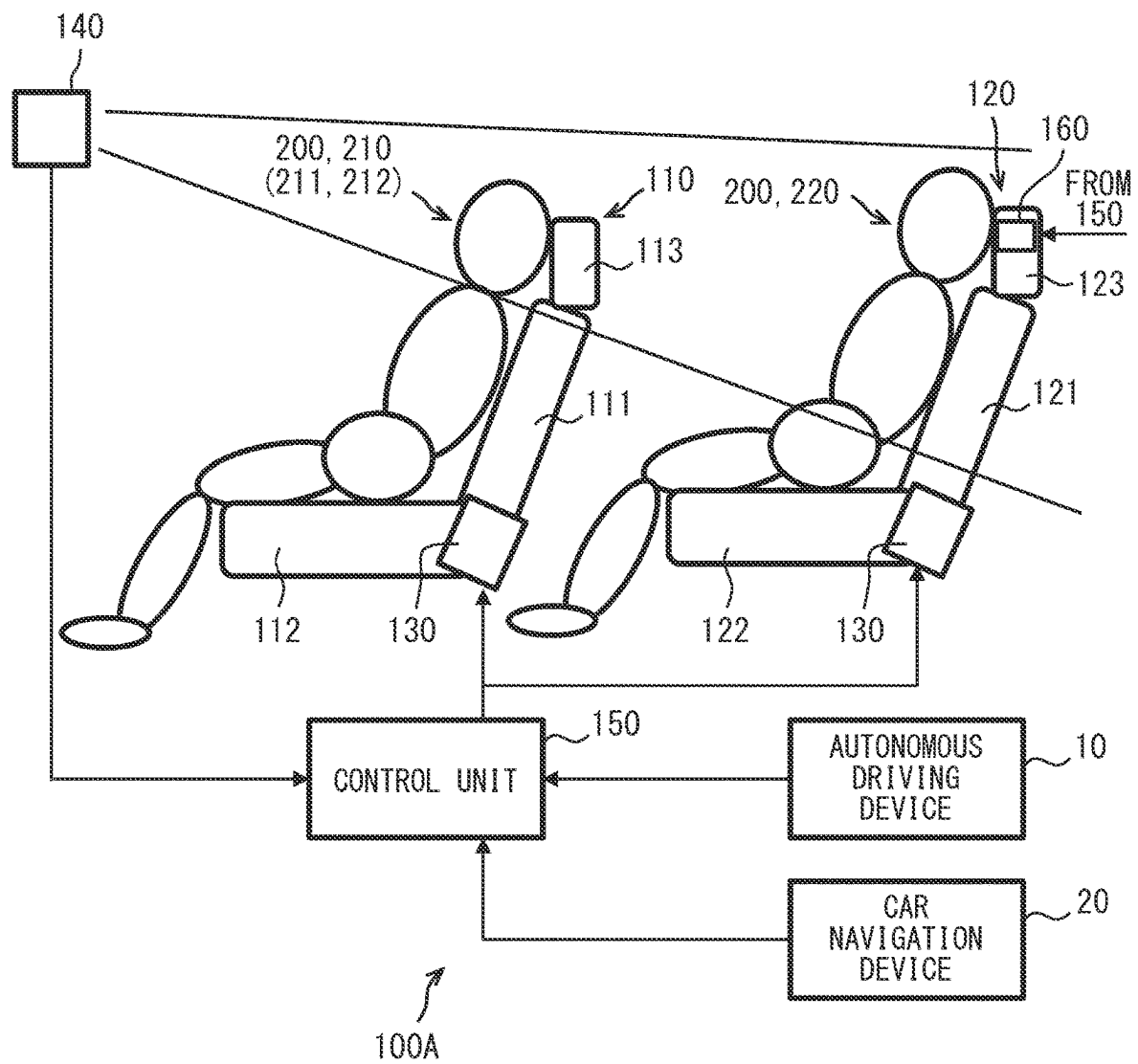
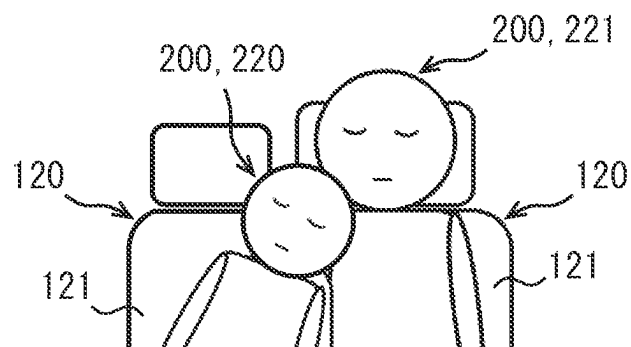


FIG. 14

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SEAT RECLINING POSITION CONTROL DEVICE

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation application of International Patent Application No. PCT/JP2021/027805 filed on Jul. 27, 2021, which designated the U.S. and claims the benefit of priority from Japanese Patent Application No. 2020-134990 filed in Japan on Aug. 7, 2020. The entire disclosures of all of the above applications are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a seat reclining position control device.

BACKGROUND

A seat reclining position control device has a detector configured to detect a drowsiness of an occupant of a vehicle based on biological information such as pulse wave data or heart rate data of the occupant. When the occupant becomes drowsy, the seat back is reclined to a position suitable for falling asleep.

SUMMARY

A seat reclining position control device includes: a seat on which an occupant of a vehicle sits; an adjustment part configured to adjust an angle of a backrest of the seat; a sleep state detection unit configured to detect a sleep state of the occupant; a control unit configured to operate the adjustment unit to control the angle; and a vehicle interior detection unit configured to detect a vehicle interior state of the vehicle. The control unit implements a seat angle control to adjust the angle of the backrest for the occupant in the sleep state based on the vehicle interior state to a suitable sleeping angle suitable for sleeping when the sleep state detection unit detects the sleep state of the occupant.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is an explanatory diagram showing a seat reclining position control device according to a first embodiment;

FIG. 2 is an explanatory diagram showing a straight posture of an occupant in the first embodiment;

FIG. 3 is an explanatory diagram showing a slightly inclined posture of an occupant;

FIG. 4 is an explanatory diagram showing a case where the occupant's head is off the headrest;

FIG. 5 is an explanatory diagram showing a case where the occupant's back is tilted forward and separated from the backrest;

FIG. 6 is an explanatory diagram showing a state before reclining in a second embodiment;

FIG. 7 is an explanatory diagram showing a state in which the backrest is reclined;

FIG. 8 is an explanatory view showing a state in which the seat portion is further moved rearward and downward;

FIG. 9 is an explanatory diagram showing a reclining state when there is no passenger on the rear seat in a third embodiment;

FIG. 10 is an explanatory diagram showing a reclining state when there is a passenger in the rear seat;

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FIG. 11 is an explanatory diagram showing a reclining state when there is a passenger in a rear seat in a fourth embodiment;

FIG. 12 is an explanatory diagram showing a seat reclining position control device according to a seventh embodiment;

FIG. 13 is an explanatory diagram showing a seat reclining position control device according to an eighth embodiment; and

FIG. 14 is an explanatory diagram showing a passenger in a rear seat and another adjacent passenger in a ninth embodiment.

DETAILED DESCRIPTION

To begin with, examples of relevant techniques will be described. A seat reclining position control device (sleep onset device) detects a drowsiness based on biological information such as pulse wave data or heart rate data of the occupant of the vehicle. When the drowsiness state is detected, the controller causes a predetermined in-vehicle device to perform an operation suitable for sleep of the occupant. As the in-vehicle device, for example, there is a seat reclining device. When the occupant becomes drowsy, the seat back is reclined to a position (angle) suitable for falling asleep.

However, the seat reclining control is performed based only on whether or not the occupant is drowsy. Nothing is taken into account, other than the drowsiness state of the occupant. A state of another occupant is not taken into consideration in the seat reclining control.

For example, depending on the posture of a drowsy occupant, reclining the seat may cause the occupant to move in an unintended direction or receive a rebound, which may cause discomfort to the occupant. Further, when there are passengers other than the drowsy occupant (to be targeted) in the rear seat, if the seat for the drowsy occupant is suddenly reclined, the other passengers feel uncomfortable.

The present disclosure provides a seat reclining position control device that can perform a seat angle control without causing discomfort to a target occupant and the other occupant.

A seat reclining position control device includes: a seat on which an occupant of a vehicle sits; an adjustment part configured to adjust an angle of a backrest of the seat; a sleep state detection unit configured to detect a sleep state of the occupant; a control unit that operates the adjustment unit to control the angle; and a vehicle interior detection unit configured to detect a vehicle interior state of the vehicle. The control unit implements a seat angle control to adjust the angle of the backrest for the occupant in the sleep state based on the vehicle interior state to a suitable sleeping angle suitable for sleeping when the sleep state detection unit detects the sleep state of the occupant.

Accordingly, the seat angle control is performed for the sleeping occupant based on the sleeping state of the occupant and the vehicle interior state detected by the vehicle interior detection unit. Therefore, it is possible to control the seat angle without giving discomfort to the sleeping occupant and the other occupant.

Hereinafter, embodiments for implementing the present disclosure will be described referring to drawings. In each embodiment, portions corresponding to the elements described in the preceding embodiments are denoted by the same reference numerals, and redundant explanation thereof may be omitted. When only a part of a configuration is described in an embodiment, the other preceding embodi-

ments can be applied to the other parts of the configuration. It may be possible not only to combine parts the combination of which is explicitly described in an embodiment, but also to combine parts of respective embodiments the combination of which is not explicitly described if any obstacle does not especially occur in combining the parts of the respective embodiments.

First Embodiment

A seat reclining position control device **100** of the first embodiment will be described with reference to FIGS. **1** to **5**. The seat reclining position control device **100** is mounted in a vehicle, for controlling an angle of a backrest **111**, **121** of a front seat **110** or a rear seat **120**, as a seat angle control, for example, for an occupant **200** in a sleeping state. The occupant **200** includes a front seat occupant **210** and a rear seat occupant **220**.

The vehicle is, for example, a five-passenger car, and the occupant is a driver **211** alone or the driver **211** and the other passengers (at least one passenger). The vehicle may include an autonomous driving device **10** and/or a car navigation device **20**.

The autonomous driving device **10** can implement automatic driving levels 1 to 5 based on the driving state of the vehicle, the driving environment (the road on which the vehicle is traveling, the surrounding environment), and the like. Automatic driving level 1 allows the autonomous driving device **10** to support either steering operation or acceleration/deceleration operation. Automatic driving level 2 allows the autonomous driving device **10** to support both the steering operation and the acceleration/deceleration operation. Automatic driving level 3 allows the autonomous driving device **10** to perform all operations in a specific place, but the driver **211** operates in an emergency. Automatic driving level 4 allows the autonomous driving device **10** to perform all operations in place of driving by the driver **211** at a specific place, and the driver **211** needs to drive after passing the specific place. Automatic driving level 5 allows the autonomous driving device **10** to perform all operations instead of driving by the driver **211**, regardless of location.

The car navigation device **20** grasps the position of the vehicle by GPS (Global Positioning System), displays the result on the map screen, and guides the user to the desired destination.

As shown in FIG. **1**, the seat reclining position control device **100** includes a front seat **110**, a rear seat **120**, an adjustment unit **130**, a camera **140**, and a control unit **150**.

The front seat **110** includes two seats for the front seat occupant **210** (such as the driver **211** and the front passenger **212**). Each front seat **110** is provided with a backrest **111**, a seat portion **112** and a headrest **113**. The front seat **110** has a reclining function of the backrest **111** and a position adjusting function of the seat portion **112**. The front seat **110** corresponds to a seat of the present disclosure.

The rear seat **120** includes three seats for the rear seat occupant **220**. Each rear seat **120** is provided with a backrest **121**, a seat portion **122** and a headrest **123**. The rear seat **120** has a function of reclining the backrest **121**. The rear seat **120** may correspond to a seat of the present disclosure. The backrest **121** of the rear seat **120** corresponds to a rear seat backrest of the present disclosure.

In the front seat **110** and the rear seat **120**, the headrest **113**, **123** can be regarded as a part of the backrest **111**, **121**.

The adjustment unit **130** is provided on each of the front seat **110** and the rear seat **120**. The adjustment unit **130** adjusts the angle of the backrest **111**, **121** (reclining func-

tion). That is, the adjustment unit **130** reclines the angle of the backrest **111**, **121** for the sleeping occupant suitably for sleeping, and restores the backrest **111**, **121** when the occupant wakes from sleep. The adjustment unit **130** also adjusts the position of the seat portion **112** of the front seat **110** in the up-down direction and the front-rear direction (position adjustment function). The adjustment unit **130** is controlled by the control unit **150** to change the angle of the backrest **111**, **121** and adjust the position of the seat portion **112**.

The camera **140** is a sleep state detection unit that detects the sleep state of the occupant **200**. The camera **140** is, for example, a CCD camera, a CMOS camera, an infrared camera, or the like, and is provided at a front side on the ceiling of the vehicle. The camera **140** acquires face images of all of the occupants **200** on the front seat **110** and the rear seat **120**. The camera **140** outputs the acquired face image data to the control unit **150**.

The camera **140** also functions as a vehicle interior detection unit that detects the state of the vehicle interior in this embodiment. The camera **140** acquires images of the seating positions and postures of all occupants **200** on board, and outputs the acquired image data (state image data) to the control unit **150**.

The control unit **150** detects an occupant in a sleeping state among the occupants **200** from the face images obtained by the camera **140**. For example, when the eyelids are continuously closed for a predetermined time or longer, the control unit **150** determines that the occupant **200** is in a sleep state from the face image of the occupant **200**. Further, the control unit **150** grasps the seating position, the posture, etc. of each occupant **200** from the state image obtained by the camera **140**. The control unit **150** grasps the current set angles of the backrests **111**, **121** and the set position of the seat portion **112** in cooperation with the adjustment unit **130**.

In general, the control unit **150** determines, as the state of the vehicle interior, relative to the seat **110**, **120**, the angle setting conditions of the backrest **111**, **121**, the position setting condition of the seat portion **112**, which seat the occupant **200** is seated in, which occupant **200** is in a sleep state, and the sitting posture of each occupant **200**.

When the vehicle is provided with the autonomous driving device **10**, the control unit **150** grasps the level (1 to 5) of automatic driving currently being executed based on the information from the autonomous driving device **10**. When the vehicle is equipped with the car navigation device **20**, the control unit **150** grasps the time required to reach the destination, the driving difficulty level of the current road, and navigation information such as highlight points before arriving at the destination based on the information from the car navigation device **20**.

Then, the control unit **150** controls the angle adjustment of the backrest **111** (or the backrest **121**) with respect to the sleeping occupant **200**, taking into account the state of the vehicle interior, the automatic driving level, the navigation information, and the like.

The configuration of the seat reclining position control device **100** of the present embodiment is as described above, and the operation and effects will be described below with reference to FIGS. **2** to **5**.

The control unit **150** detects an occupant **200** in a sleep state in the front seat **110** or the rear seat **120** from the face images of the occupants **200** obtained from the camera **140**.

The driver **211** can be included in the occupant **200** in a sleep state. When the driver **211** is in a sleep state, for example, the vehicle is controlled by the autonomous driv-

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ing device **10** at the automatic driving level 4 or 5, or the vehicle is parked or stopped in a parking lot while the driver is taking a nap. It is desired that the driver **211** be preferentially relaxed during an automatic driving or during a nap, for safe driving. The occupant **200** in a sleep state includes a passenger **212** such as rear seat occupant **220**, other than the driver **211**.

The control unit **150** detects the posture of the sleeping occupant **200** from the state image of the occupant **200** obtained from the camera **140**. As shown in FIGS. 2 and 3, for example, when the sleeping occupant **200** is not out of the backrest **111** and/or the headrest **113**, the control unit **150** reclines the backrest **111** to adjust the angle of the backrest **111** to a sleeping angle suitable for sleeping (as seat angle control). The sleeping angle of the backrest **111** is set in advance for sleeping. In case where the sleeping occupant **200** is a rear seat occupant **220**, the seat angle control is performed to the backrest **121**.

As shown in FIG. 4, when the head of the sleeping occupant **200** is off the headrest **113**, the control unit **150** does not execute the seat angle control. Further, the seat angle control is not executed (prohibited) when the upper body is tilted forward as shown in FIG. 5 since the back of the occupant **200** is not in contact with the backrest **111**.

As described above, in the present embodiment, the control unit **150** performs the seat angle control based on the sleeping state and the seating posture of the occupant **200**. As described with reference to FIG. 4, if the seat angle control is performed when the head of the occupant **200** is out of the headrest **113**, the occupant **200** may move in an unintended direction (slip down from the seat). As described with reference to FIG. 5, if the seat angle control is performed when the back of the occupant **200** is not in contact with the backrest **111**, the back of the occupant **200** may hit the backrest **111** by receiving a recoil due to reclining. In this embodiment, considering the above, the seat angle control is performed when the occupant **200** is not separated from the backrest **111** and/or the headrest **113**, so that the occupant **200** does not feel uncomfortable.

When the sleeping occupant **200** is the driver **211** and the vehicle is automatically traveling, or when the driver **211** is taking a nap while the vehicle is parked, the control unit **150** controls the seat angle for the driver **211**. Therefore, the driver **211** can have a comfortable sleep.

Second Embodiment

A seat angle control in the second embodiment will be described with reference to FIGS. 6 to 8. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment. In the second embodiment, a seat angle control is performed by utilizing that the seat portion **112** of the front seat **110** is adjustable (movable) in position. The directions in which the position of the seat portion **112** can be adjusted are the up-down direction and the front-rear direction.

The sleeping occupant **200** is herein assumed to be the front seat occupant **210** on the front seat **110**, for example, the driver **211** or the front passenger **212** (FIG. 6).

The control unit **150** performs the seat angle control shown in FIG. 7 when there is no rear seat occupant **220** (another passenger) in the rear seat **120** at rear side of the sleeping occupant **200**, as the vehicle interior state. In addition, as shown in FIG. 8, the control unit **150** moves the seat portion **112** rearward and downward.

This makes it easier for the sleeping occupant **200** to stretch his or her legs, thereby enabling sleep in a more

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comfortable environment. Since there is no occupant **200** in the rear seat **120**, there is no rear seat occupant **220** feeling uncomfortable.

Third Embodiment

A seat angle control in the third embodiment will be described with reference to FIGS. 9 and 10. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment. In the third embodiment, when there is a rear seat occupant **220** (another passenger) in the rear seat **120** at the rear side of the sleeping occupant **200** (front seat occupant **210**) as a vehicle interior state, the control unit **150** reduces the reclining angle shown in FIG. 10, to be smaller than the normal sleeping angle shown in FIG. 9 when performing the seat angle control for the sleeping occupant **210** (front seat occupant **210**).

When the control unit **150** determines that the front seat occupant **210** in a sleeping state has sufficient foot space (there is a predetermined space or more) from the control position of the seat portion **112**, the control unit **150** may perform the seat angle control and move the position of the seat portion **112** forward.

Accordingly, the seat angle control for the front seat occupant **210** can be performed without impairing the comfort of the rear seat occupant **220** (so as not to make the rear seat occupant **220** uncomfortable).

Fourth Embodiment

FIG. 11 shows the seat angle control in the fourth embodiment. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment. In the fourth embodiment, when there is a rear seat occupant **220** (another passenger) in the rear seat **120** on the rear side of the sleeping occupant **200** (front seat occupant **210**) as a vehicle interior state, the control unit **150** performs the seat angle control for the sleeping occupant **210**, and, as shown in FIG. 11, reclines the backrest **121** (rear seat backrest) of the rear seat **120**.

When the rear seat occupant **220** is also in a sleeping state, the backrest **121** of the rear seat **120** is reclined at the same angle as the backrest **111** of the front seat **110** (FIG. 11). When the rear seat occupant **220** is awake, the reclining angle of the backrest **121** of the rear seat **120** is adjusted to be smaller than the normal sleeping angle.

Accordingly, the seat angle control for the front seat occupant **210** can be performed without impairing the comfort of the rear seat occupant **220** (so as not to make the rear seat occupant **220** uncomfortable). When the sleeping occupant **200** is the driver **211**, the seat angle control (such as full reclining) is possible to prioritize the driver **211** more than in case of the third embodiment.

Fifth Embodiment

In the fifth embodiment, a seat angle control is implemented while the vehicle is automatically driven at the automatic driving level 4 or 5 by the autonomous driving device **10** and when the destination guidance is performed by the car navigation device **20**. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment.

The control unit **150** grasps the required time to the destination from the navigation information of the car navigation device **20**. When the time required to reach the destination is shorter than a predetermined time, the control

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unit **150** prohibits the seat angle control for the driver **211**, and permits the seat angle control for the other passengers (the passenger seat occupant **212**, the rear seat occupant **220**) other than the driver **211**. Since the optimum time for a nap is usually 15 to 30 minutes, the predetermined time (determination time) can be set to, for example, about 5 to 10 minutes.

As a result, when the required time (distance) to the destination is short and sufficient sleep time cannot be secured, the driver **211** can be prevented from falling asleep. The other passengers can sleep comfortably by performing the seat angle control.

Sixth Embodiment

In the sixth embodiment, the procedure for adjusting the seat angle when driving on a road with a difficulty level during automatic driving will be described. The difficulty level of driving is high along, for example, city roads, roads with frequent vehicle movement, mountain roads with a series of curves, cliff roads, and the like. The control unit **150** acquires road information including the driving difficulty level from the car navigation device **20**. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment.

The control unit **150** prohibits the seat angle control for the driver **211**, and permits the seat angle control for the other passengers (the passenger seat occupant **212**, the rear seat occupant **220**) other than the driver **211**, when the vehicle is automatically traveling on a road with a predetermined driving difficulty level.

As a result, even during automatic driving, by prohibiting the seat angle control for the driver **211** in case a danger occurs, preparations are made so that the driver can control the driving by himself/herself at any time. The other passengers can sleep comfortably by implementing the seat angle control.

Seventh Embodiment

FIG. **12** shows a seat reclining position control device **100A** according to the seventh embodiment. The seat reclining position control device **100A** includes a notification unit **160** for notifying the rear seat occupant **220**. The notification unit **160** can be, for example, a directional speaker that has directivity to each rear seat occupant **220** on the rear seat **120**. The notification unit **160** is provided in, for example, the headrest **123**.

When the control unit **150** performs the seat angle control, if a rear seat occupant **220** (another passenger) is in the rear seat **120** at the rear side of the sleeping occupant **200** (the driver **211** or the front passenger **212**) and is awake, the notification unit **160** notifies the rear seat occupant **220** in advance that the seat angle control will be performed. Specifically, the notification unit **160** notifies that, for example, "be careful as the front seat will be reclined." If the rear seat occupant **220** is in a sleep state, the control unit **150** prohibits the notification to the rear seat occupant **220**.

As a result, the seat angle control can be performed without startling the rear seat occupant **220** due to the sudden seat reclining (so as not to give discomfort) by giving advance notification.

Eighth Embodiment

FIG. **13** shows a seat reclining position control device **100B** according to the eighth embodiment. The seat reclining

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position control device **100B** includes an awakening unit **170** for awakening the sleeping occupant **200**. The awakening unit **170** can be, for example, a vibration generator that generates vibrations to the sleeping occupant **200**, a volume adjuster that increases the volume of an audio device, or the like. When the awakening unit **170** is set as a vibration generator, the awakening unit **170** is provided, for example, in the backrest **111** as shown in FIG. **13**. When the awakening unit **170** is set as a volume adjuster, the awakening unit **170** is provided, for example, in an audio device.

If the sleeping occupant **200** is the driver **211** and the vehicle is automatically traveling, the control unit **150** controls the seat angle for the driver **211** and helps for awakening the driver **211** when the time until the end of the automatic driving becomes shorter than a predetermined time, for example, using the above-described vibration generator of volume adjuster (awakening unit **170**).

As a result, when the driver **211** is sleeping during automatic driving, the driver **211** can be awakened before the automatic driving ends, and the automatic driving can be transitioned smoothly to the manual driving by the driver **211**.

Ninth Embodiment

FIG. **14** shows the seat angle control in the ninth embodiment. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment. In the ninth embodiment, when the control unit **150** detects the sleeping occupant **200** and an adjacent occupant **221** adjacent to the sleeping occupant **200**, the control unit **150** performs the seat angle control for the sleeping occupant **200** and the adjacent occupant **221** in the same manner.

In this embodiment, the sleeping occupant **200** is a rear seat occupant **220** (for example, a child) seated in the center of the rear seat **120**, and the adjacent occupant **221** (for example, a parent) is seated next to the rear seat occupant **220**.

When the adjacent occupant **221** is in a sleep state as well as the rear seat occupant **220**, the control unit **150** sets the angle of the backrest **121** of the adjacent occupant **221** to be the same as the angle of the backrest **121** of the rear seat occupant **220**. When the adjacent occupant **221** is not sleeping, the control unit **150** may set the angle of the backrest **121** of the adjacent occupant **221** to be smaller than the angle of the backrest **121** of the rear seat occupant **220**.

As a result, the backrest **121** for the adjacent occupant **221** can be reclined in the same manner as for the rear seat occupant **220**, thereby reducing the occurrence of a step between the backrests **121**, so as to improve convenience (by reducing discomfort).

Tenth Embodiment

In the tenth embodiment, when the seat angle control is performed for the sleeping occupant **200**, the reclining angle is returned to the original setting according to the navigation information from the car navigation device **20**. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment.

The navigation information may indicate that the vehicle will reach a predetermined highlight point (such as a scenic point on the way, a famous place, or the destination itself) when the vehicle is being guided to the destination. When this navigation information is obtained, the control unit **150** restores the angle of the backrest **111** or the backrest **121** for the occupant **200** by cancelling the seat angle control.

As a result, it is possible to wake up the occupant **200** and prevent from overlooking the highlight points during the driving by returning the backrest **111**, **121** to the normal position.

Eleventh Embodiment

In the eleventh embodiment, the control unit **150** prohibits the seat angle control when the occupant **200** has already changed the angle of the backrest **111** or the backrest **121** while detecting that the occupant **200** is in a sleep state. The configuration of the seat reclining position control device **100** is the same as that of the first embodiment.

If the occupant **200** himself/herself changes the angle of the backrest **111** or the backrest **121**, the angle is likely to be the angle that the occupant **200** prefers. Therefore, by prohibiting the seat angle control, the angle set by the occupant **200** can be maintained.

OTHER EMBODIMENT

In the embodiments, the relationship between the front seat occupant **210** and the rear seat occupant **220** is described using the front seat **110** and the rear seat **120** in the seat angle control. However, for example, when a vehicle has three rows of seats, such as a minivan, it is possible to use the second row of seats as the front seat **110** and the third row of seats as the rear seat **120**.

In the embodiments, the camera **140** is used as the sleep state detection unit, but is not limited to this. The sleep state detection unit may be a biosensor to detect the sleeping state from the heartbeat, pulse rate, or the like of the occupant **200**.

In the embodiments, the camera **140** for detecting the sleep state is also used as the vehicle interior detection unit, but a dedicated camera may be used.

The control unit and the techniques thereof according to the present disclosure may be implemented by one or more special-purposed computers. Such a special-purposed computer may be provided by configuring a processor and a memory programmed to execute one or more functions embodied by a computer program.

Alternatively, the control unit or the method disclosed may be provided by a dedicated computer which includes at least one processor configured by at least one dedicated hardware logic circuits.

Alternatively, the control unit and the like and the method thereof described in the present disclosure may be achieved by one or more dedicated computers constituted by a combination of a processor and a memory programmed to execute one or a plurality of functions and a processor constituted by one or more hardware logic circuits.

The computer program may be stored in a computer-readable non-transition tangible storage medium as an instruction executed by a computer.

Although the present disclosure has been described in accordance with the embodiments, it is understood that the present disclosure is not limited to such embodiments or structures. The present disclosure also encompasses various modifications and variations within the scope of equivalents. In addition, while the various elements are shown in various combinations and configurations, which are exemplary, other combinations and configurations, including more, less or only a single element, are also within the spirit and scope of the present disclosure.

What is claimed is:

1. A seat reclining position control device comprising:
 - a seat on which an occupant of a vehicle sits;
 - an adjustment unit configured to adjust an angle of a backrest of the seat;
 - a sleep state detection unit configured to detect a sleep state of the occupant;
 - a control unit configured to operate the adjustment unit to control the angle; and
 - a vehicle interior detection unit configured to detect a vehicle interior state of the vehicle, wherein the control unit implements a seat angle control to adjust the angle of the backrest for the occupant in a sleep state based on the vehicle interior state to a sleeping angle suitable for sleeping when the sleep state detection unit detects that the occupant is in the sleep state, and the vehicle is provided with an autonomous driving device that enables an automatic driving, and a car navigation device that grasps a position of the vehicle and performs a destination guidance, in response to determining that the vehicle is traveling automatically and in response to determining that a time required to reach a destination by the destination guidance is shorter than a predetermined time, the control unit prohibits an execution of the seat angle control for a driver and permits an execution of the seat angle control for a passenger other than the driver.
2. The seat reclining position control device according to claim 1, wherein
 - the control unit obtains a posture of the occupant in the sleep state as the vehicle interior state, and
 - the control unit implements the seat angle control when the occupant is not separated from the backrest.
3. The seat reclining position control device according to claim 1, wherein
 - the adjustment unit is able to adjust a position of a seat portion of the seat, and
 - the control unit implements the seat angle control and moves the position of the seat portion rearward and downward when there is no passenger in a rear seat at a rear of the occupant in the sleep state as the vehicle interior state.
4. The seat reclining position control device according to claim 1, wherein
 - when another occupant is in a rear seat at a rear of the occupant in the sleep state as the vehicle interior state, the control unit reduces the angle to be smaller than the sleeping angle for the occupant in the sleep state in performing the seat angle control.
5. The seat reclining position control device according to claim 1, wherein
 - when another occupant is in a rear seat at a rear of the occupant in the sleep state as the vehicle interior state, the control unit implements the seat angle control for the occupant in the sleep state and reclines a rear seatback of the rear seat.
6. The seat reclining position control device according to claim 1, wherein in response to determining that the vehicle is traveling automatically along a road with a predetermined driving difficulty level, the control unit prohibits the execution of the seat angle control for the driver and permits the execution of the seat angle control for the passenger other than the driver.
7. The seat reclining position control device according to claim 1, further comprising: a notification unit configured to notify the occupant, wherein

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in case where another occupant is in a rear seat at a rear of the occupant in the sleep state in performing the seat angle control,

when the another occupant is awake, the control unit controls the notification unit to notify the another occupant that the seat angle control will be performed in advance, and

when the another occupant is in a sleeping state, the control unit prohibits the notification unit from notifying the another occupant.

8. The seat reclining position control device according to claim 1, further comprising:

an awakening unit configured to awaken the occupant in the sleep state,

in case where the occupant in the sleep state is the driver, the control unit implements the seat angle control for the driver when the vehicle is automatically traveling, and the control unit activates the awakening unit when a time until an end of the automatic driving becomes shorter than a predetermined time.

9. The seat reclining position control device according to claim 1, wherein

when the control unit detects the occupant in the sleep state and an adjacent occupant adjacent to the occupant in the sleep state, the control unit implements the seat angle control for both of the occupant in the sleep state and the adjacent occupant.

10. The seat reclining position control device according to claim 1, wherein

after the seat angle control is implemented for the occupant in the sleep state, the control unit restores the angle to an original setting before the vehicle reaches a predetermined highlight point.

11. The seat reclining position control device according to claim 1, wherein the control unit prohibits an execution of the seat angle control when the angle has already been changed by the occupant even while detecting the occupant in the sleep state.

12. A seat reclining position control device comprising: a computer including a processor and a memory that stores instructions configured to, when executed by the processor, cause the processor to

receive at least one of a sleep state of an occupant of a vehicle or a vehicle interior state of the vehicle from a camera, and

implement a seat angle control to adjust an angle of a seat for the occupant in a sleep state based on the sleep state or/and the vehicle interior state, wherein

the processor prohibits an execution of the seat angle control for a driver and permits an execution of the seat angle control for a passenger other than the driver, in response to an autonomous driving device causing the vehicle to travel automatically and a car navigation

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device determining that a time required for the vehicle to reach a destination is shorter than a predetermined time.

13. The seat reclining position control device according to claim 12, wherein

the processor prohibits the execution of the seat angle control for the driver and permits the execution of the seat angle control for the passenger other than the driver, in response to the autonomous driving device causing the vehicle to travel automatically along a road with a predetermined driving difficulty level.

14. The seat reclining position control device according to claim 12, wherein

the processor implements the seat angle control for the driver of the vehicle that is the occupant in the sleep state when the autonomous driving device causes the vehicle to travel automatically, and

the processor activates an awakening unit configured to awaken the occupant in the sleep state when a time until an end of automatic driving becomes shorter than a predetermined time.

15. The seat reclining position control device according to claim 12, wherein

after the seat angle control is implemented for the occupant in the sleep state, the processor restores the angle to an original setting before the vehicle reaches a predetermined highlight point while the car navigation device performs a destination guidance.

16. The seat reclining position control device according to claim 12, wherein the processor prohibits an execution of the seat angle control when the angle has already been changed by the occupant even while detecting the occupant in the sleep state.

17. The seat reclining position control device according to claim 12, wherein

the processor implements the seat angle control and moves a position of a seat portion of the seat rearward and downward when there is no passenger in a rear seat at a rear of the occupant in the sleep state as the vehicle interior state.

18. The seat reclining position control device according to claim 12, wherein

when detecting another occupant in a rear seat at a rear of the occupant in the sleep state as the vehicle interior state, the processor reduces the angle to be smaller than a sleeping angle for the occupant in the sleep state in performing the seat angle control.

19. The seat reclining position control device according to claim 12, wherein

when detecting another occupant in a rear seat at a rear of the occupant in the sleep state as the vehicle interior state, the processor implements the seat angle control for the occupant in the sleep state and reclines a rear seatback of the rear seat.

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